

# Retrieval-Based Adaptation of Time Series Foundation Models

Experiment Guideline

Gaspard Berthelier

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# 1 Time Series Forecasting

Let  $\mathcal{U} = \{0, \dots, U-1\}$  be a set of  $U$  users or channels and  $t \in \{0, \dots, T-1\}$  the time index for the  $T$  measured dates. For channel  $u \in \mathcal{U}$ , the univariate series is  $z_0^{(u)}, \dots, z_{T-1}^{(u)}$ . We consider the setting of a fixed lookback length  $L \geq 1$  and forecast horizon  $H \geq 1$ . Given a valid *query date*  $s$ , the lookback and future horizon are

$$X_{s,u} = z_{(s-L,s]}^{(u)} = (z_{s-L+1}^{(u)}, \dots, z_s^{(u)}) \in \mathbb{R}^L, \quad (1)$$

$$Y_{s,u} = z_{(s,s+H]}^{(u)} = (z_{s+1}^{(u)}, \dots, z_{s+H}^{(u)}) \in \mathbb{R}^H. \quad (2)$$

Thus the lookback length  $L$  represents the number of observed values at a given time and the horizon  $H$  is the number of target values to predict.

A forecasting model is a parameterized map

$$F_\theta : \mathbb{R}^L \longrightarrow \mathbb{R}^H, \quad X_{s,u} \longmapsto \hat{Y}_{s,u}.$$

Here  $\theta$  denotes the model parameters.

Certain families of models allow the use of optional covariates. Let  $C_{s,u}$  denote a set of covariate windows of size  $L+H$  (horizon included). When using covariates, the prediction becomes  $\hat{Y}_{s,u}^{\text{cov}} = F_\theta(X_{s,u}; C_{s,u})$ .

For any  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ , define

$$\mu_x = \frac{1}{n} \sum_{k=1}^n x_k, \quad \sigma_x = \text{sd}(x) = \left( \frac{1}{n} \sum_{k=1}^n (x_k - \mu_x)^2 \right)^{1/2}, \quad \varepsilon = 10^{-8}.$$

For one sample, consisting of windows  $(X, Y, \hat{Y})$ , the principal losses are

$$\text{MSE}(\hat{Y}, Y) = \frac{1}{H} \sum_{h=1}^H (\hat{Y}_h - Y_h)^2, \quad \text{nMSE}(\hat{Y}, Y \mid X) = \frac{1}{H} \sum_{h=1}^H \left( \frac{\hat{Y}_h - Y_h}{\max(\sigma_X, \varepsilon)} \right)^2.$$

## 2 Foundation Models

A time series Foundation Model (FM) is pretrained over many tasks and datasets, and directly reused on unseen series without updating its parameters. Indeed, their extensive training allows them to perform In-Context Learning (ICL), which means the model has learned to detect patterns within the input, as if it were a dataset on which it was fine-tuned. A few examples of FMs are provided below.

**Chronos-2.** Chronos-2 is a 120M-parameter encoder-only foundation model whose group attention supports univariate, multivariate, and covariate-aware zero-shot forecasting (Ansari et al., 2025). In this project, we notably use context mode `future_included`: each covariate contributes its lookback as well as its observed horizon. The active code name is `chronos2`.

**TabPFN-TS.** TabPFN-TS casts forecasting as tabular regression with time, seasonal, frequency, and optional covariate features, then performs ICL with a pretrained TabPFN regressor (Hoo et al., 2025). The project wrapper uses the `tabpfm-v2.5-regressor-v2.5_default.ckpt` checkpoint, automatically adds periods 24 and 168 when  $L$  permits, and exposes the code name `tabpfnts`.

**TS-ICL.** TS-ICL is a 27M-parameter time-indexed encoder-regressor Transformer for forecasting and imputation on regular or irregular timestamps (Le Naour et al., 2026). It is part of the literature scope, but `ts_icl` is deliberately rejected by the current launcher because no wrapper is registered.

**Chronos-Bolt.** The `chronos-bolt` wrapper supplies the frozen 205M-parameter Chronos-Bolt-base backbone for the Cross-RAG comparison. It is univariate and ignores retrieved covariates; neighbour targets and forecasts remain available to the adapter (Ansari et al., 2024; Amazon Science, 2026). Source-faithful local model modules load the released base weights from `weights/chronos-bolt-base` and the sole recursive `best.pth` below `weights/cross-rag`; `CHRONOS_BOLT_WEIGHTS_PATH` and `CROSSRAG_WEIGHTS_PATH` are explicit overrides. No external code checkout or Cross-RAG retraining is part of the workflow.

Table 1: Model names present in the Python registry or publication launchers. Trainable means parameters updated by this project, not total pretrained size.

| Code name           | Role and current status                           | Total parameters              | Trainable here |
|---------------------|---------------------------------------------------|-------------------------------|----------------|
| <b>chronos2</b>     | Primary screen/full backbone; frozen              | 120M                          | 0              |
| <b>tabpfnts</b>     | Ultra backbone; frozen TabPFN-v2.5 regressor      | $\approx 11$ M                | 0              |
| <b>chronos-bolt</b> | Fixed Cross-RAG backbone; frozen                  | 205M                          | 0              |
| <b>persistence</b>  | Python-registry diagnostic                        | 0                             | 0              |
| <b>linear</b>       | Python-registry diagnostic, univariate            | $H(L + 1)$                    | $H(L + 1)$     |
| <b>patchtst</b>     | Registry diagnostic; not launched for publication | 466,843 / 663,091 / 1,762,603 | same           |
| <b>ts_icl</b>       | Reserved shell name; not registered               | 27M in the paper              | not run        |

The three PatchTST numbers in Table 1 use the code defaults at 168:24, 336:48, and 504:168, respectively. Publication experiments accept only the three frozen backbones listed first.

### 3 From RAG to retrieval-augmented forecasting

Retrieval-augmented generation separates a parametric model from an explicit, replaceable non-parametric memory. Classical RAG retrieves documents from a dense index and conditions a pretrained generator on them (Lewis et al., 2020). In  $k$ NN-MT, a frozen neural machine-translation model caches decoder representations  $k_i$  with next-token values  $v_i$ ; at decoding state  $q_t$ , it interpolates the model and neighbour distributions (Khandelwal et al., 2021). Let  $p_\theta(\cdot | q_t)$  be the neural next-token distribution for candidate token  $y_t$ , let  $\mathcal{N}_K(q_t)$  contain the  $K \geq 1$  cached keys nearest to  $q_t$  under distance  $d$ , and choose interpolation weight  $\lambda \in [0, 1]$  and temperature  $\tau > 0$ . Then

$$p(y_t | q_t) = (1 - \lambda)p_\theta(y_t | q_t) + \lambda \frac{\sum_{i \in \mathcal{N}_K(q_t)} \mathbf{1}[v_i = y_t] \exp(-d(q_t, k_i)/\tau)}{\sum_{i \in \mathcal{N}_K(q_t)} \exp(-d(q_t, k_i)/\tau)}.$$

This is the direct ancestor of our question: can a frozen pretrained model be adapted cheaply by retrieving domain examples and mixing their outcomes?

Time-series work has followed several paths: retrieved sequences can become a text prompt (TimeRAG), an appended context (RAF), an input to a model trained on the target dataset (RAFT), or values fused by pretrained attention modules (TS-RAG and Cross-RAG). Our method is narrower: the datastore is target-panel history, the backbone is frozen, and simple validated forecasts or gates are tested before a neural adapter.

Table 2: Bibliography map for pretrained retrieval and time-series forecasting. This is a method map, not a cross-paper accuracy ranking.

| Method                              | Date / venue               | Retrieval object                                    | Adaptation mechanism                                                                         |
|-------------------------------------|----------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------|
| RAG (Lewis et al., 2020)            | 2020, NeurIPS              | Dense-indexed passages                              | Fine-tuned seq2seq generator combines parametric and non-parametric memory.                  |
| $k$ NN-MT (Khandelwal et al., 2021) | 2021, ICLR                 | Decoder state / next-token pairs                    | No retraining; interpolates neural and neighbour token distributions.                        |
| RAF (Tire et al., 2024)             | 2024 preprint; v2 2026     | Related time-series contexts and futures            | Normalises, aligns, and prepends retrieved examples to frozen TSFMs.                         |
| TimeRAG (Yang et al., 2025)         | 2024 preprint; ICASSP 2025 | DTW-similar historical sequences                    | Renders references and query as an LLM forecasting prompt.                                   |
| TS-RAG (Ning et al., 2025a)         | 2025, NeurIPS              | Encoder-space context-future pairs                  | Pretrains projectors and an Adaptive Retrieval Mixer; no target-task tuning.                 |
| RAFT (Han et al., 2025)             | 2025, ICML                 | Correlated training inputs and following patches    | Trains a shallow multiresolution forecasting MLP per target dataset.                         |
| Cross-RAG (Lee et al., 2026a)       | 2026, KDD MILETS           | Min-max lookback/future pairs, cosine search        | Pretrained query-retrieval cross-attention with query-dependent and query-independent paths. |
| This project                        | 2026                       | Raw or transformed historical lookback/future pairs | Closed-form baselines, CatBoost gates, and TS-IFA around a frozen backbone.                  |

## 4 Neighbour retrieval

### 4.1 Datastore and exact search

Fix a named retrieval transform  $\rho$  and distance metric  $\delta$ . Their maps are

$$\phi = \phi_\rho : \mathbb{R}^L \rightarrow \mathbb{R}^{d_\rho}, \quad d = d_\delta : \mathbb{R}^{d_\rho} \times \mathbb{R}^{d_\rho} \rightarrow \mathbb{R}_{\geq 0},$$

with every implemented  $\phi_\rho$  and  $d_\delta$  defined explicitly in Table 3. Here  $d_\rho = L$  except for the encoder space;  $g_\theta : \mathbb{R}^L \rightarrow \mathbb{R}^{d_{\text{enc}}}$  denotes the frozen backbone representation after the configured flattening or pooling.

For query  $(s, u)$ , let  $\mathcal{R}_s = \{r_{s,0} < \dots < r_{s,n_s^{\text{store}}-1}\}$  be the ordered finite set of  $n_s^{\text{store}} = |\mathcal{R}_s|$  store dates in  $\{L-1, \dots, T-H-1\}$  that pass the fixed/online, period, stride, and cap rules in Section 7. For scope  $q \in \{\text{all, same, other}\}$ , the eligible store keys are

$$\mathcal{I}_{s,u}^{\text{all}} = \mathcal{R}_s \times \mathcal{U}, \quad \mathcal{I}_{s,u}^{\text{same}} = \mathcal{R}_s \times \{u\}, \quad \mathcal{I}_{s,u}^{\text{other}} = \mathcal{R}_s \times (\mathcal{U} \setminus \{u\}).$$

The labelled datastore  $\mathcal{D}_{s,u}^q$  contains  $(\phi(X_{r,v}), X_{r,v}, Y_{r,v}, r, v)$  for every  $(r, v) \in \mathcal{I}_{s,u}^q$ . For  $K \in \{1, \dots, |\mathcal{I}_{s,u}^q|\}$ , let  $\text{TopK}^{\min}$  return the ordered  $K$  minimizers. Exact search returns the following store keys:

$$\mathcal{N}_K^q(s, u) = \text{TopK}_{(r,v) \in \mathcal{I}_{s,u}^q}^{\min} d(\phi(X_{s,u}), \phi(X_{r,v})).$$

The datastore is flattened in *user-major, date-minor* order. For  $i \in \{0, \dots, Un_s^{\text{store}} - 1\}$ , a flat index decodes as

$$v = \left\lfloor \frac{i}{n_s^{\text{store}}} \right\rfloor, \quad m = i \bmod n_s^{\text{store}}, \quad r = r_{s,m}.$$

Downstream query payloads use the opposite, date-major ordering  $(s_0, u_0), \dots, (s_0, u_{U-1}), (s_1, u_0), \dots$  so that whole dates form contiguous blocks for chronological splitting;  $s_0 < s_1 < \dots$  are the saved query dates and  $u_k = k$  is the internal channel index.

In Table 3,  $X \in \mathbb{R}^L$  is a generic lookback; in its distance rows,  $a, b \in \mathbb{R}^{d_\rho}$  are two transformed lookbacks and  $x_{\min} = \min_{1 \leq k \leq L} X_k$ ,  $x_{\max} = \max_{1 \leq k \leq L} X_k$ ,  $a^\circ = a - \mu_a \mathbf{1}_{d_\rho}$ ,  $b^\circ = b - \mu_b \mathbf{1}_{d_\rho}$  are their centred forms.

Table 3: Implemented retrieval spaces and distances.

| Space     | $\phi(X)$                                                                                          | Status               |
|-----------|----------------------------------------------------------------------------------------------------|----------------------|
| Raw       | $X$                                                                                                | Primary screen       |
| Instance  | $(X - \mu_X \mathbf{1}_L) / (\sigma_X + \varepsilon)$                                              | Primary screen       |
| Min-max   | $(X - x_{\min} \mathbf{1}_L) / \max(x_{\max} - x_{\min}, \varepsilon)$                             | Cross-RAG profile    |
| Fourier   | $ \text{FFT}((X - \mu_X \mathbf{1}_L) / (\sigma_X + \varepsilon)) $                                | Optional ablation    |
| Encoder   | flattened or pooled frozen representation $g_\theta(X)$                                            | Implemented override |
| Euclidean | $\ a - b\ _2$                                                                                      | Primary metric       |
| Cosine    | $1 - a^\top b / [\max(\ a\ _2, \varepsilon) \max(\ b\ _2, \varepsilon)]$                           | Cross-RAG metric     |
| Pearson   | $1 - (a^\circ)^\top b^\circ / [\max(\ a^\circ\ _2, \varepsilon) \max(\ b^\circ\ _2, \varepsilon)]$ | Implemented override |

### 4.2 Saved information and query-scale transfer

For neighbour rank  $j \in \{1, \dots, K\}$  with  $(r_j, v_j) \in \mathcal{N}_K(s, u)$ , abbreviate  $X_j = X_{r_j, v_j}$  and  $Y_j = Y_{r_j, v_j}$ , and let  $d_j$  be its retrieval distance and  $N_j = F_\theta(X_j)$  its frozen forecast. Also abbreviate the query as  $X = X_{s,u}$  and  $Y = Y_{s,u}$ . Extraction saves the quantities in Table 4; it stores  $E_j = Y_j - N_j$  rather than duplicating  $N_j$ , which is reconstructed downstream.

Table 4: Current extraction payload. Optional fields are generated only when their switches are enabled.

| Object     | Saved content                                                           |
|------------|-------------------------------------------------------------------------|
| Query      | $X, Y, V = F_\theta(X), C = F_\theta(X; \{X_j, Y_j\}_{j=1}^K)$          |
| Neighbours | $X_j, Y_j, E_j = Y_j - N_j, d_j, r_j, v_j$ for $j = 1, \dots, K$        |
| Indices    | query date/user, neighbour date/user, eligible store-date/window counts |
| Summaries  | query and neighbour level/scale; forecast-context and residual losses   |
| Optional   | context-conditioned neighbour residuals                                 |

Retrieved values are expressed on the query scale before fitting. Write  $q = (s, u)$  for the current query and let  $(\mu_q, \sigma_q) = (\mu_X, \sigma_X)$  and  $(\mu_j, \sigma_j) = (\mu_{X_j}, \sigma_{X_j})$ . For a neighbour-scale horizon vector  $Z \in \mathbb{R}^H$  or residual vector  $E \in \mathbb{R}^H$ , define

$$\mathcal{T}_{j \rightarrow q}(Z) = \frac{Z - \mu_j \mathbf{1}_H}{\max(\sigma_j, \varepsilon)} \sigma_q + \mu_q \mathbf{1}_H, \quad (3)$$

$$\mathcal{T}_{j \rightarrow q}^{\text{res}}(E) = \frac{E}{\max(\sigma_j, \varepsilon)} \sigma_q. \quad (4)$$

All later symbols  $Y_j, N_j, E_j$  denote these query-scale values.

Distances are converted to weights independently for every query. Let  $\mathbf{d} = (d_1, \dots, d_K)$  be the ordered retrieved-distance vector and let  $j, \ell \in \{1, \dots, K\}$ :

$$\tilde{d}_j = \frac{d_j - \min_{\ell} d_{\ell}}{\max(\sigma_{\mathbf{d}}, \varepsilon)}, \quad w_j = \frac{e^{-\tilde{d}_j}}{\sum_{\ell=1}^K e^{-\tilde{d}_{\ell}}}, \quad \bar{Y}_h = \sum_{j=1}^K w_j Y_{j,h}.$$

Thus  $w_j \geq 0$ ,  $\sum_{j=1}^K w_j = 1$ , and  $\bar{Y} \in \mathbb{R}^H$ ; for  $K = 1$ ,  $w_1 = 1$ .

## 5 Retrieval baselines

For extracted window  $i$  with query  $(s_i, u_i)$ , define the target  $G_i = Y_{s_i, u_i} \in \mathbb{R}^H$ , vanilla forecast  $V_i \in \mathbb{R}^H$ , conditioned forecast  $C_i \in \mathbb{R}^H$ , query-scale neighbour future and forecast  $(Y_{i,j}, N_{i,j}) \in \mathbb{R}^H \times \mathbb{R}^H$  for  $j \in \{1, \dots, K\}$ , and

$$V_i = F_{\theta}(X_{s_i, u_i}), \quad C_i = F_{\theta}(X_{s_i, u_i}; \{X_{i,j}, Y_{i,j}\}_{j=1}^K), \quad \bar{Y}_i = \sum_{j=1}^K w_{i,j} Y_{i,j}.$$

Each design is  $(V_i, Z_{i,1}, \dots, Z_{i,p})$ .

Table 5: Baseline designs;  $Z_i = (Z_{i,1}, \dots, Z_{i,p})$ .

| Stem     | $Z_i$                                                     | $p$      |
|----------|-----------------------------------------------------------|----------|
| cov      | $(C_i)$                                                   | 1        |
| avgy     | $(\bar{Y}_i)$                                             | 1        |
| y        | $(Y_{i,1}, \dots, Y_{i,K})$                               | $K$      |
| cov_y    | $(C_i, Y_{i,1}, \dots, Y_{i,K})$                          | $1 + K$  |
| cov_avgy | $(C_i, \bar{Y}_i)$                                        | 2        |
| residual | $(Y_{i,1}, \dots, Y_{i,K}, N_{i,1}, \dots, N_{i,K})$      | $2K$     |
| full     | $(C_i, Y_{i,1}, \dots, Y_{i,K}, N_{i,1}, \dots, N_{i,K})$ | $1 + 2K$ |

Let  $\tau \in \{S, H\}$  denote shared or horizon-wise coefficients. For any coefficient vector  $c$ , set  $c_h^S = c$  for every  $h$ , whereas  $c_1^H, \dots, c_H^H$  are fitted independently. Define the simplex  $\mathcal{S}_p = \{\lambda \in \mathbb{R}^{p+1} : \lambda_m \geq 0, \sum_{m=0}^p \lambda_m = 1\}$ . The three families are

$$\hat{G}_{i,h}^{R,\tau} = V_{i,h} + a_h^{\tau} V_{i,h} + \sum_{m=1}^p b_{h,m}^{\tau} Z_{i,h,m}, \quad (a_h^{\tau}, b_{h,1}^{\tau}, \dots, b_{h,p}^{\tau}) \in \mathbb{R}^{p+1}, \quad (5)$$

$$\hat{G}_{i,h}^{C,\tau} = \lambda_{h,0}^{\tau} V_{i,h} + \sum_{m=1}^p \lambda_{h,m}^{\tau} Z_{i,h,m}, \quad \lambda_h^{\tau} \in \mathcal{S}_p, \quad (6)$$

$$\hat{G}_{i,h}^{D,\tau} = V_{i,h} + \sum_{m=1}^p b_{h,m}^{\tau} (Z_{i,h,m} - V_{i,h}), \quad b_h^{\tau} \in \mathbb{R}^p. \quad (7)$$

Vanilla is recovered by  $(a, b) = 0$ ,  $\lambda = (1, 0, \dots, 0)$ , or  $b = 0$ . Ridge contains every convex forecast through  $a = \lambda_0 - 1$  and  $b_m = \lambda_m$ . For cov, the formulas are  $V + aV + bC$ ,  $\lambda V + (1 - \lambda)C$ , and  $V + b(C - V)$ .

Let  $\mathcal{Q}_A$  be the windows in split  $A \in \{T_1, T_2, T_{12}\}$ , where  $T_{12} = T_1 \cup T_2$ . For  $\tau = S$ , set  $\mathcal{O}_A^S = \mathcal{Q}_A \times \{1, \dots, H\}$ ; for the  $h$ th horizon fit, set  $\mathcal{O}_A^{H,h} = \mathcal{Q}_A \times \{h\}$ . Ridge and delta-ridge use correction target  $t_{i,h} = G_{i,h} - V_{i,h}$  and design

$$d_{i,h}^R = (V_{i,h}, Z_{i,h,1}, \dots, Z_{i,h,p}) \in \mathbb{R}^{p+1}, \quad d_{i,h}^D = (Z_{i,h,1} - V_{i,h}, \dots, Z_{i,h,p} - V_{i,h}) \in \mathbb{R}^p.$$

For either  $d \in \mathbb{R}^q$  and an observation set  $\mathcal{O}$ , define

$$r_k = \max \left( \sqrt{\frac{1}{|\mathcal{O}|} \sum_{(i,h) \in \mathcal{O}} d_{i,h,k}^2}, 10^{-12} \right), \quad \tilde{d}_{i,h,k} = d_{i,h,k}/r_k,$$

and, for  $\alpha \geq 0$ ,

$$\hat{\gamma}_{\alpha, \mathcal{O}} = \arg \min_{\gamma \in \mathbb{R}^q} \frac{1}{|\mathcal{O}|} \sum_{(i,h) \in \mathcal{O}} (t_{i,h} - \tilde{d}_{i,h}^\top \gamma)^2 + \alpha \|\gamma\|_2^2, \quad \hat{\beta}_{\alpha, \mathcal{O}, k} = \hat{\gamma}_{\alpha, \mathcal{O}, k} / r_k.$$

The grid is

$$\mathcal{A} = \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}.$$

For each  $\alpha \in \mathcal{A}$ , coefficients fitted on  $T_1$  are scored by nMSE on  $T_2$ ; the minimizer (smallest  $\alpha$  on ties) is refitted on  $T_{12}$ . Convex coefficients are fitted directly on  $T_{12}$ :

$$\hat{\lambda}_{\mathcal{O}} = \arg \min_{\lambda \in \mathcal{S}_p} \frac{1}{|\mathcal{O}|} \sum_{(i,h) \in \mathcal{O}} \left( G_{i,h} - \lambda_0 V_{i,h} - \sum_{m=1}^p \lambda_m Z_{i,h,m} \right)^2.$$

Use  $\mathcal{O} = \mathcal{O}_A^S$  for a shared fit and  $\mathcal{O} = \mathcal{O}_A^{H,h}$  independently for each  $h$ ; no intercept or centring is added. Chunked sufficient statistics give the exact objectives.

Table 6: Baseline names; substitute a stem from Table 5.

| Artifact             | Definition                       | Fitted scalars   |
|----------------------|----------------------------------|------------------|
| vanilla              | $V_{i,h}$                        | 0                |
| cov_forecast         | $C_{i,h}$                        | 0                |
| avgy                 | $\bar{Y}_{i,h}$                  | 0                |
| y_mean               | $K^{-1} \sum_j Y_{i,j,h}$        | 0                |
| <stem>_ridge_*       | Equation (5), * = shared/horizon | $(p+1) / H(p+1)$ |
| <stem>_convex_*      | Equation (6), * = shared/horizon | $(p+1) / H(p+1)$ |
| <stem>_delta_ridge_* | Equation (7), * = shared/horizon | $p / Hp$         |

## 6 Gates and features

A gate compares  $V_i$  with  $A_i = C_i$  (cov) or  $A_i = \bar{Y}_i$  (avgy). Define the pointwise and shared candidate advantages

$$\Delta_{i,h} = (G_{i,h} - V_{i,h})^2 - (G_{i,h} - A_{i,h})^2, \quad \bar{\Delta}_i = H^{-1} \sum_{h=1}^H \Delta_{i,h}.$$

For  $q = (q_1, \dots, q_K) \in \mathbb{R}^K$ , define neighbour-weighted moments

$$\mu_{w_i}(q) = \sum_{j=1}^K w_{i,j} q_j, \quad \sigma_{w_i}(q) = \left[ \sum_{j=1}^K w_{i,j} (q_j - \mu_{w_i}(q))^2 \right]^{1/2}.$$

Let  $\mu_h$  and  $\sigma_h$  denote population moments over  $h \in \{1, \dots, H\}$ . For neighbour  $j$ , set

$$\begin{aligned} r_{i,j}^V &= \mu_h(Y_{i,j,h} - V_{i,h}), & s_{i,j}^V &= \sigma_h(Y_{i,j,h} - V_{i,h}), \\ r_{i,j}^N &= \mu_h(Y_{i,j,h} - N_{i,j,h}), & s_{i,j}^N &= \sigma_h(Y_{i,j,h} - N_{i,j,h}). \end{aligned}$$

Let  $(r_{i,j}, v_{i,j})$  be neighbour  $j$ 's date and channel, and define

$$\begin{aligned} \ell_i &= (\mu_{X_{i,1}}, \dots, \mu_{X_{i,K}}), & \rho_i &= (\sigma_{X_{i,1}}, \dots, \sigma_{X_{i,K}}), & \mathbf{a}_i &= (s_i - r_{i,1}, \dots, s_i - r_{i,K}), \\ \mathbf{w}_i &= (w_{i,1}, \dots, w_{i,K}), & \mathbf{d}_i &= (d_{i,1}, \dots, d_{i,K}). \end{aligned}$$

The exact static vector  $f_i^0 \in \mathbb{R}^{18}$  is

$$\begin{aligned} f_i^0 &= (\mu_{w_i}(\mathbf{r}_i^V), \sigma_{w_i}(\mathbf{r}_i^V), \mu_{w_i}(\mathbf{s}_i^V), \mu_{w_i}(\mathbf{r}_i^N), \sigma_{w_i}(\mathbf{r}_i^N), \mu_{w_i}(\mathbf{s}_i^N), \\ &\quad \mu_{X_{s_i, u_i}}, \sigma_{X_{s_i, u_i}}, \mu_{\ell_i}, \sigma_{\ell_i}, \mu_{\rho_i}, \sigma_{\rho_i}, K^{-1} \sum_{j=1}^K \mathbf{1}[v_{i,j} = u_i], \\ &\quad \mu_{\mathbf{a}_i}, \sigma_{\mathbf{a}_i}, \sigma_{\mathbf{w}_i}, \max_j w_{i,j}, \mu_{\mathbf{d}_i}), \end{aligned}$$

where  $\mathbf{r}_i^V = (r_{i,j}^V)_{j=1}^K$  and analogously for  $\mathbf{s}_i^V, \mathbf{r}_i^N, \mathbf{s}_i^N$ . The shared and horizon feature vectors are

$$f_i^S = (\mu_h(A_{i,h} - V_{i,h}), \sigma_h(A_{i,h} - V_{i,h}), f_i^0) \in \mathbb{R}^{20}, \quad (8)$$

$$f_{i,h}^H = (A_{i,h} - V_{i,h}, \mu_{w_i}((Y_{i,j,h} - V_{i,h})_j), \sigma_{w_i}((Y_{i,j,h} - V_{i,h})_j), \mu_{w_i}((Y_{i,j,h} - N_{i,j,h})_j), \sigma_{w_i}((Y_{i,j,h} - N_{i,j,h})_j), f_i^0) \in \mathbb{R}^{23}. \quad (9)$$

For shape  $\tau \in \{S, H\}$ , write  $z_i^S = \bar{\Delta}_i$ ,  $z_{i,h}^H = \Delta_{i,h}$ ,  $f_i^S$  as in Equation (8), and  $f_{i,h}^H$  as in Equation (9). A shared model is fitted once; a horizon model is fitted independently for each  $h$ . CatBoost scores are

$$g^{\text{reg},\tau} = \text{CBReg}(f^\tau), \quad g^{\text{cls},\tau} = \Pr_{\text{CBCls}}(z^\tau > 0 \mid f^\tau) - \frac{1}{2}.$$

The tree count is selected on  $T_2$  after fitting on  $T_1$ , then the model is refitted on  $T_{12}$  with that count. Define

$$q^{\text{hard},\tau} = \mathbf{1}[g^\tau > 0], \quad \hat{G}_{i,h}^{\text{hard},\tau} = (1 - q_{i,h}^{\text{hard},\tau})V_{i,h} + q_{i,h}^{\text{hard},\tau}A_{i,h},$$

with  $q_{i,h}^{\cdot,S} = q_i^{\cdot,S}$ . For soft output, let  $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$  and

$$s^S = \text{sd}((z_i^S)_{i \in \mathcal{Q}_{T_{12}}}), \quad s_h^H = \text{sd}((z_{i,h}^H)_{i \in \mathcal{Q}_{T_{12}}}).$$

Writing  $s^\tau$  for the matching shared or  $h$ th-horizon scale,

$$p^{\text{cls},\tau} = g^{\text{cls},\tau} + \frac{1}{2}, \quad p^{\text{reg},\tau} = \begin{cases} \text{sigmoid}(g^{\text{reg},\tau}/s^\tau), & s^\tau > 10^{-12}, \\ 0, & s^\tau \leq 10^{-12}, g^{\text{reg},\tau} < 0, \\ \frac{1}{2}, & s^\tau \leq 10^{-12}, g^{\text{reg},\tau} = 0, \\ 1, & s^\tau \leq 10^{-12}, g^{\text{reg},\tau} > 0, \end{cases}$$

$$\hat{G}_{i,h}^{\text{soft},\tau} = (1 - p_{i,h}^\tau)V_{i,h} + p_{i,h}^\tau A_{i,h}.$$

For shared output,  $p_{i,h}^{\cdot,S} = p_i^{\cdot,S}$  for every  $h$ . The no-feature Bayes scores and scored-split oracle scores are

$$g^{B,S} = |\mathcal{Q}_{T_{12}}|^{-1} \sum_{i \in \mathcal{Q}_{T_{12}}} \bar{\Delta}_i, \quad g_h^{B,H} = |\mathcal{Q}_{T_{12}}|^{-1} \sum_{i \in \mathcal{Q}_{T_{12}}} \Delta_{i,h}, \quad g_i^{O,S} = \bar{\Delta}_i, \quad g_{i,h}^{O,H} = \Delta_{i,h},$$

and use the hard rule above.

Table 7: Gate names for  $A \in \{\text{cov}, \text{avgy}\}$ ;  $\tau = S/H$ .

| Name pattern                           | Definition                   | Fitted scalars / proxy     |
|----------------------------------------|------------------------------|----------------------------|
| bayes_<A>_shared/horizon               | $g^{B,\tau}$ , hard          | 1 / $H$                    |
| catboost_<A>_regressor_shared/horizon  | $g^{\text{reg},\tau}$ , hard | $\leq 6000$ / $\leq 6000H$ |
| catboost_<A>_classifier_shared/horizon | $g^{\text{cls},\tau}$ , hard | $\leq 6000$ / $\leq 6000H$ |
| catboost_<A>_*_shared/horizon_soft     | same fit, soft output        | 0 additional               |
| oracle_<A>_shared/horizon              | $g^{O,\tau}$ , hard          | 0                          |

CatBoost uses at most 300 depth-4 trees, learning rate 0.03, 50-round early stopping, and balanced classifier classes (Prokhorenkova et al., 2018). The proxy is  $300(16 \text{ leaf} + 4 \text{ split}) = 6000$ . Screen uses only hard shared regressors; classifier, soft-regressor, and horizon-regressor forms are one-axis ablations.

## 7 Practical protocol

### 7.1 Chronological regions and window eligibility

For the  $T$  dates introduced in Section 1, extraction computes exclusive boundaries, where round means nearest-integer rounding and  $[a, b) = \{a, a+1, \dots, b-1\}$  for integer dates:

$$b_0 = \text{round}(0.30T), \quad b_{12} = \text{round}(0.80T), \quad b_3 = T.$$

The extraction regions are the fixed datastore region  $T_0 = [0, b_0)$ , pooled adaptation region  $T_{12} = [b_0, b_{12})$ , and held-out evaluation region  $T_3 = [b_{12}, T)$ . For a target period  $[b, e)$ , eligible query dates are

$$s \in [\max(L-1, b-1), \min(T-H-1, e-H-1)]$$

at the configured query stride. Hence the first target may start at  $b$  with query index  $b - 1$ ; its lookback may cross the boundary, but its complete target must be inside  $[b, e)$ . Changing  $L$  therefore preserves  $T_3$  target dates whenever both runs remain eligible.

Each downstream learner re-splits  $T_{12}$  by unique query date. If  $n_{12} \geq 2$  dates are present and the validation fraction is  $\rho = 0.2$ ,

$$n_2 = \min(n_{12} - 1, \max(1, \text{round}(\rho n_{12}))), \quad n_1 = n_{12} - n_2.$$

The first  $n_1$  and last  $n_2$  whole dates form  $T_1$  and  $T_2$ , respectively, so  $T_{12} = T_1 \cup T_2$  and the two sets are disjoint; users on the same date are never split.

## 7.2 Datastore date grids

Let  $P$  be the retrieval period,  $D_{\text{stride}} > 0$  the datastore stride, and  $A \in \{0, 1\}$  the period-alignment indicator. Alignment requires  $P \mid D_{\text{stride}}$ . Before optional explicit history bounds, define the admissible endpoint pair

$$(a_s, b_s) = \begin{cases} (L - 1, b_0 - H - 1), & \text{fixed mode,} \\ (L - 1, s - H), & \text{online mode.} \end{cases}$$

The first candidate and uncapped date grid are

$$r_s^A = \begin{cases} a_s + ((s - a_s) \bmod P), & A = 1, \\ a_s, & A = 0, \end{cases} \quad \mathcal{G}_s = \{r_s^A + \ell D_{\text{stride}} : \ell \in \mathbb{N}_0, r_s^A + \ell D_{\text{stride}} \leq b_s\}.$$

Let  $\text{Tail}_M(\mathcal{G})$  denote the  $M$  most recent ordered elements of  $\mathcal{G}$ , with  $\text{Tail}_{+\infty}(\mathcal{G}) = \mathcal{G}$ . Let  $M_s$  be the minimum of the active date cap, the window-derived cap  $\lfloor W_{\text{max}}/U \rfloor$ , and, in default online mode, the number of dates in the fixed-mode grid for the same  $s$ ; an absent cap is  $+\infty$ , and `full_online_history` omits the last term. The code returns

$$\mathcal{R}_s = \text{Tail}_{M_s}(\mathcal{G}_s).$$

Thus  $A = 1$  implies  $(s - r) \bmod P = 0$  for every  $r \in \mathcal{R}_s$ ;  $A = 0$  imposes no phase constraint. In fixed mode, both  $(r - L, r]$  and  $(r, r + H]$  lie in  $T_0$ . In online mode,

$$r \geq L - 1, \quad r + H \leq s,$$

so the neighbour future is already observed. It may overlap the query lookback, but never the query target.

Without an explicit override, the code first counts the dates that a fixed  $T_0$  store would provide under the same value of  $A$ . The online store then keeps that many *most recent* eligible dates, subject to the 30,000-window cap. Thus the store slides forward without growing merely because the query is later. `full_online_history` removes this implicit date-count cap; explicit date and window caps still apply. Defaults are  $P = 96$  samples for ETTm1, ETTm2, ETT\_T\_15T, and ETT\_L\_15T, and  $P = 24$  otherwise. Publication profiles disable period alignment so their datastore origins rotate through seasonal phases. If period alignment is explicitly enabled, the datastore stride must be a multiple of  $P$ .

## 7.3 Fitting and leakage controls

Table 8: What each chronological region may influence.

| Family                   | $T_1$                                     | $T_2$                                 | $T_3$                            |
|--------------------------|-------------------------------------------|---------------------------------------|----------------------------------|
| Convex mixtures          | diagnostic fit                            | final fit on pooled $T_{12}$          | score once                       |
| Ridge baselines          | fit all $\alpha$                          | select $\alpha$ ; refit pooled        | score once                       |
| Fixed-candidate CatBoost | fit up to 300 trees                       | select tree count; fresh pooled refit | score once                       |
| TS-IFA joint             | optimize branches and one rooter          | select and restore checkpoint         | score once                       |
| TS-IFA meta              | chronological support/query outer updates | fit active frozen-branch rooter       | score once                       |
| Oracle diagnostics       | —                                         | —                                     | target-aware appendix bound only |

Ridge sufficient statistics are accumulated exactly in float64 chunks. Optional sample caps affect fitting only; final  $T_3$  scoring always uses every saved sample. A future gate over a trainable candidate must train that candidate on  $T_1$  and the gate on  $T_2$  without pooled refitting, unless out-of-fold candidate predictions are introduced.

Dataset loaders discover `config.json` beside the selected CSV. Portable fields live at the JSON top level and adaptation-only overrides under `adaptation`. Project-scoped values override shared values, explicit run arguments override both, and `drop_users` is additive. The resolved path and applied keys are recorded in the extraction manifest; no file content hash is created.

## 8 TS-IFA

TS-IFA is a lightweight neural adapter around the frozen backbone  $F := F_\theta$ . We again suppress  $(s, u)$ , so  $X = X_{s,u}$  and  $Y = Y_{s,u}$ ; all following values are on the query scale:

$$p = F(X), \quad p^c = F(X; \{X_j, Y_j\}_{j=1}^K), \quad n_j = F(X_j), \quad e_j = Y_j - n_j.$$

### 8.1 Candidate branches

Define the learned maps

$$\begin{aligned} A_r &: \mathbb{R}^{L+H} \times (\mathbb{R}^{L+H})^K \times (\mathbb{R}^H)^K \rightarrow \mathbb{R}^H, & M_r &: \mathbb{R}^{2H} \rightarrow \mathbb{R}^H, \\ A_m &: \mathbb{R}^L \times (\mathbb{R}^L)^K \times (\mathbb{R}^H)^K \rightarrow \mathbb{R}^H, & M_m &: \mathbb{R}^{2H} \rightarrow \mathbb{R}^H. \end{aligned}$$

$A_r, A_m$  are multi-head cross-attention maps and  $M_r, M_m$  are correction MLPs. Residual attention compares query and neighbour *states*:

$$q_r = [X, p], \quad k_j^r = [X_j, n_j], \quad v_j^r = e_j, \quad (10)$$

$$z_r = A_r(q_r, \{k_j^r\}, \{v_j^r\}), \quad c_r = p + M_r([p, z_r]). \quad (11)$$

Memory attention reads realised neighbour horizons directly:

$$z_m = A_m(X, \{X_j\}, \{Y_j\}), \quad c_m = p + M_m([p, z_m]).$$

Vanilla is always present, and any non-empty subset of  $\{p^c, c_r, c_m\}$  can be selected. For active candidate set  $I$ , the unconstrained rooters operate on the non-vanilla deltas  $d_j = c_j - p$ ,  $j \in I$ . Fixed and learned transformed covariates are a separate experiment rather than TS-IFA candidates.

### 8.2 Neural and ridge rooters

The neural roter receives no handcrafted distances or dispersion features. Its learned attention map is  $A_g : \mathbb{R}^L \times (\mathbb{R}^L)^K \times (\mathbb{R}^L)^K \rightarrow \mathbb{R}^{d_g}$ , where  $d_g$  is the configured representation width:

$$z_g = A_g(X, \{X_j\}, \{X_j\}) \in \mathbb{R}^{d_g}.$$

For the learned type embedding  $\tau_j \in \mathbb{R}^{d_g}$  of active delta  $d_j$ , let LN denote dimension-preserving layer normalization and let  $S_\omega : \mathbb{R}^{2d_g+2H} \rightarrow \mathbb{R}^R$  be the shared MLP scorer, where  $R = 1$  for shared routing and  $R = H$  for horizon routing. It returns

$$a_j = S_\omega([\text{LN}(z_g), \text{LN}(p), \text{LN}(d_j), \text{LN}(\tau_j)]) \in \mathbb{R}^R, \quad (12)$$

$$\hat{Y}_{\text{neural}} = p + \sum_{j \in I} a_j \odot d_j. \quad (13)$$

Here  $\omega$  denotes all neural-roter parameters:  $A_g$ , its layer normalizations, the type embeddings, and the shared scorer. Unconstrained routing uses these signed delta weights. Optional softmax routing prepends a vanilla logit, then normalizes vanilla and every active candidate to non-negative weights summing to one. The shared form broadcasts one value over all horizons; the horizon form retains one value per horizon index.

The ridge roter uses the same active candidates. In the joint variant its free routing matrix is  $\nu \in \mathbb{R}^{|I| \times R}$  and is updated by gradient descent with the branches. Only the meta variant uses the following closed-form solve. For any

non-empty fit set  $\mathcal{A}$ , let  $d_{i,j,h}$  be the horizon- $h$  delta of candidate  $j$  on window  $i$ . Let  $n_{\mathcal{A}} = |\mathcal{A}| \geq 1$ . The differentiable meta-solve uses  $\epsilon_{\text{meta}} = 10^{-8}$  and the final fit uses  $\epsilon_{\text{fit}} = 10^{-12}$ :

$$s_{j,h}^{(d,\mathcal{A})} = \begin{cases} \sqrt{n_{\mathcal{A}}^{-1} \sum_{i \in \mathcal{A}} d_{i,j,h}^2 + \epsilon_{\text{meta}}^2}, & \text{differentiable meta-solve,} \\ \max\left(\sqrt{n_{\mathcal{A}}^{-1} \sum_{i \in \mathcal{A}} d_{i,j,h}^2}, \epsilon_{\text{fit}}\right), & \text{final fit,} \end{cases} \quad \tilde{d}_{i,j,h} = d_{i,j,h} / s_{j,h}^{(d,\mathcal{A})}.$$

For ridge penalty  $\alpha > 0$ , the standardized coefficient matrix is

$$\Gamma_{\mathcal{A}}^*(\theta_b) = \arg \min_{\Gamma \in \mathbb{R}^{3 \times H}} \sum_{i \in \mathcal{A}} \sum_{h=1}^H \left( Y_{i,h} - p_{i,h} - \sum_{j=1}^3 \Gamma_{j,h} \tilde{d}_{i,j,h} \right)^2 + \alpha \|\Gamma\|_F^2, \quad \nu_{\mathcal{A},j,h}^*(\theta_b) = \Gamma_{\mathcal{A},j,h}^*(\theta_b) / s_{j,h}^{(d,\mathcal{A})},$$

where  $\|\cdot\|_F$  is the Frobenius norm. This exact solve applies to unconstrained meta-ridge; softmax meta-ridge uses differentiable gradient inner updates. The unconstrained ridge prediction is  $p_{i,h} + \sum_{j=1}^3 \nu_{j,h} d_{i,j,h}$ . Both ridge variants have exactly  $|I|R$  routing parameters;  $\alpha = 10^{-2}$  applies to exact meta-ridge solves only.

### 8.3 Training and initialization

Let  $\theta_b$  collect the parameters of  $c_r$  and  $c_m$ , and let  $\mathcal{B} = \{c_r, c_m\}$ . For minibatch index set  $\mathcal{A}$ , active-rooter parameters  $\psi \in \{\nu, \omega\}$ , and coefficients  $a_{i,j,h}$ , define

$$\mathcal{L}_B^{\mathcal{A}}(\theta_b) = \sum_{c \in \mathcal{B}} \text{nMSE}_{\mathcal{A}}(c, Y) + \lambda_V \sum_{c \in \mathcal{B}} \text{nMSE}_{\mathcal{A}}(c, p), \quad (14)$$

$$\mathcal{L}_R^{\mathcal{A}}(\theta_b, \psi) = \text{nMSE}_{\mathcal{A}}(\hat{Y}, Y) + \lambda_V \text{nMSE}_{\mathcal{A}}(\hat{Y}, p) + \frac{\lambda_2}{3|\mathcal{A}|H} \sum_{i,j,h} a_{i,j,h}^2 + \frac{\lambda_S}{3|\mathcal{A}|(H-1)} \sum_{i,j,h < H} (a_{i,j,h+1} - a_{i,j,h})^2. \quad (15)$$

The last term is zero for  $H = 1$ . Active values are  $\lambda_V = \lambda_2 = \lambda_S = 10^{-2}$  and  $\lambda_B = 0.1$ .

The two joint variants use all of  $T_1$ :

$$(\theta_b^*, \nu^*) = \arg \min_{\theta_b, \nu} \mathcal{L}_R^{T_1}(\theta_b, \nu) + \lambda_B \mathcal{L}_B^{T_1}(\theta_b), \quad \text{joint ridge,} \quad (16)$$

$$(\theta_b^*, \omega^*) = \arg \min_{\theta_b, \omega} \mathcal{L}_R^{T_1}(\theta_b, \omega) + \lambda_B \mathcal{L}_B^{T_1}(\theta_b), \quad \text{joint neural.} \quad (17)$$

AdamW updates both parameter blocks. Joint ridge never uses a closed-form solve. The minimum-nMSE complete checkpoint on  $T_2$  is restored, then scored once on  $T_3$ ; no joint rooter is refitted on  $T_2$ .

For meta learning, split  $T_1$  by whole query dates into earlier support  $\mathcal{S}_1$  and later query  $\mathcal{Q}_1$ . Meta ridge computes

$$\nu_e^*(\theta_b) = \nu_{\mathcal{S}_1^{(e)}}^*(\theta_b), \quad (18)$$

$$\theta_b^* = \arg \min_{\theta_b} \mathcal{L}_R^{\mathcal{Q}_1^{(e)}}(\theta_b, \nu_e^*(\theta_b)) + \lambda_B \mathcal{L}_B^{\mathcal{Q}_1^{(e)}}(\theta_b), \quad (19)$$

differentiating through each exact standardized  $|I| \times |I|$  solve. It freezes  $\theta_b^*$  and fits  $\nu_{T_2}^*(\theta_b^*)$  in closed form. Meta neural starts from learned  $\omega^{(0)} = \omega$  and takes

$$\omega^{(k+1)} = \omega^{(k)} - \eta_{\text{in}} \nabla_{\omega^{(k)}} \mathcal{L}_R^{\mathcal{S}_1^{(e)}}(\theta_b, \omega^{(k)}).$$

Its query loss updates  $\theta_b$  and the initialization  $\omega$ ; first-order MAML is the default. It then freezes  $\theta_b$  and fits only  $\omega$  on all  $T_2$ . Both meta variants score  $T_3$  once. The default query fraction is 0.2,  $K_{\text{in}} = 1$ , and  $\eta_{\text{in}} = 10^{-3}$ .

Output-zero initialization of  $M_r$ ,  $M_m$ , the neural scorer, and the free joint-ridge matrix gives  $c_r = c_m = p$ ,  $a_j = 0$ , and  $\hat{Y} = p$  at step zero. All AdamW stages use learning rate  $10^{-5}$ , weight decay  $10^{-4}$ , batch size 256, and 20,000 one-random-batch epochs outside smoke mode. The exact variant, split, optimizer, regularizers, widths, caps, and seed enter the completion signature.

### 8.4 Parameter counts

The full horizon-routing reference configuration uses four attention heads, per-head dimension 32, 128-unit hidden layers, and three rooter deltas. Shared routing or a branch subset reduces the rooter and branch counts.  $K$  affects compute and memory, not parameter count.

Table 9: Full horizon-routing TS-IFA reference parameter counts, reproduced by the implementation and synchronized `config.json` artifacts.

| $L$ | $H$ | Branches | Neural router | Full neural | Branches + ridge |
|-----|-----|----------|---------------|-------------|------------------|
| 168 | 24  | 522,080  | 323,576       | 845,656     | 522,152          |
| 336 | 48  | 645,056  | 397,424       | 1,042,480   | 645,200          |
| 504 | 168 | 915,872  | 508,616       | 1,424,488   | 916,376          |
| 512 | 64  | 759,808  | 471,232       | 1,231,040   | 760,000          |

At 512:64, the default ridge router has  $3H = 192$  coefficients.

Table 10: Adaptation-parameter comparison at the released Cross-RAG 512:64 setting. Frozen backbone parameters are excluded from the first numeric column. External counts follow the released implementations (Ning et al., 2025b; Lee et al., 2026b).

| Method                               | Trainable adaptation params. | $\times$ TS-IFA | Share of backbone           |
|--------------------------------------|------------------------------|-----------------|-----------------------------|
| TS-IFA neural                        | 1,231,040                    | 1.00            | 1.03% of Chronos-2          |
| TS-IFA branches + ridge              | 760,000                      | 0.62            | 0.63% of Chronos-2          |
| TS-RAG added modules                 | 4,775,425                    | 3.88            | 2.33% of Bolt-base          |
| Cross-RAG added modules              | 8,712,192                    | 7.08            | 4.25% of Bolt-base          |
| Full horizon ridge baseline, $K = 3$ | 512                          | 0.00042         | 0.00025% of Bolt-base       |
| One shared CatBoost gate             | $\leq 6,000$ proxy           | $\leq 0.0049$   | $\leq 0.005\%$ of Chronos-2 |

## 9 Experimental design

### 9.1 Datasets and model profiles

Table 11: Dataset panels and their roles. All non-date variates are retained for ETT, Weather, and Exchange.

| Role                    | Datasets                                   | Preparation                                                                                                                              |
|-------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Primary screen          | Electricity, Traffic, Solar, exchange_rate | Hourly Electricity/Traffic; Solar is summed from 10-minute to hourly; Exchange is daily. Electricity drops the nine configured channels. |
| Full homogeneous panels | ETT_T_1H, ETT_L_1H, ETT_T_15T, ETT_L_15T   | Temperature and load are separated. Hourly panels use $P = 24$ ; 15-minute panels use $P = 96$ .                                         |
| Mixed-quantity ablation | ETTh1, ETTh2, ETTm1, ETTm2, Weather        | Original seven-variable ETT panels; Weather is hourly mean aggregation. Cross-variable retrieval is reported only as an ablation.        |

Table 12: Resolved experiment profiles. A selected winner name fixes family, formula, retrieval space, metric,  $K$ , and online mode.

| Profile               | Datasets                     | $(L, H)$                   | Backbone                | Retrieval / methods                                                                       |
|-----------------------|------------------------------|----------------------------|-------------------------|-------------------------------------------------------------------------------------------|
| <b>test</b>           | Electricity                  | 504:168                    | Chronos-2               | raw Euclidean, $K = 3$ ; smoke-only                                                       |
| <b>screen</b>         | four primary                 | 168:24, 336:48,<br>504:168 | Chronos-2               | raw/instance Euclidean, $K \in \{1, 3\}$ ; 10 baselines and 8 gate/reference artifacts    |
| <b>full</b>           | primary + four separated ETT | default three              | Chronos-2               | selected complete screen pipelines only                                                   |
| <b>ultra</b>          | same as full                 | default three              | Chronos-2, TabPFN-TS    | same winners and output root as full                                                      |
| <b>sota_benchmark</b> | seven paper datasets         | 512:64                     | Chronos-Bolt            | selected method on exact paper test splits plus immutable published TS-RAG/Cross-RAG rows |
| <b>tsrag</b>          | four primary                 | 512:64                     | Chronos-2, Chronos-Bolt | fixed Chronos-T5 Euclidean retrieval at $K = 10$ ; released TS-RAG is Bolt-only           |

Let

$$\mathcal{D}_P = \{\text{Electricity, Traffic, Solar, exchange\_rate}\}, \quad \mathcal{S} = \{168:24, 336:48, 504:168\},$$

$$\mathcal{R} = \{\text{raw, instance}\} \times \{\text{Euclidean}\} \times \{K = 1, 3\} \times \{\text{online}\},$$

and let  $\mathcal{D}_B$  be the seven stems in Table 5. The  $12 = |\mathcal{D}_P||\mathcal{S}|$  screen settings test, for every  $r \in \mathcal{R}$ ,

$$\mathcal{B}_{\text{screen}} = \{\text{cov\_forecast, avgy, y\_mean}\} \cup \{\text{<d>_ridge\_shared} : d \in \mathcal{D}_B\},$$

$$\mathcal{G}_{\text{screen}} = \bigcup_{A \in \{\text{cov, avgy}\}} \{\text{direct\_<A>, bayes\_<A>\_shared, catboost\_<A>\_regressor\_shared, oracle\_<A>\_shared}\},$$

where **direct\_cov**=**cov\_forecast** and **direct\_avgy**=**avgy**. Thus the baseline and gate screens contain 10 and 8 artifacts per retrieval pipeline; convex, delta, horizon, classifier, and mixture forms are excluded.

Default publication datastore, pooled-adaptation-query, and evaluation-query strides are 25/25/127, respectively, with period alignment disabled. All three strides are coprime with the 24-step hourly and 96-step quarter-hourly periods, so sampled origins rotate through every seasonal phase. The datastore cap is 30,000 windows. The test profile uses aligned strides 168/256/256 and 2,048 store windows. Seed 1 is used throughout.

## 9.2 Benchmarks and ablations

Table 13: Selected baseline controls and one-family transformations. Retrieval metric is Euclidean and mode is online in every row.

| Stem            | Space, $K$  | Horizon front          | Convex front          | Delta front          |
|-----------------|-------------|------------------------|-----------------------|----------------------|
| <b>full</b>     | instance, 3 | ridge shared / horizon | ridge / convex shared | ridge / delta shared |
| <b>full</b>     | raw, 1      | ridge shared / horizon | ridge / convex shared | ridge / delta shared |
| <b>cov_y</b>    | instance, 3 | ridge shared / horizon | ridge / convex shared | ridge / delta shared |
| <b>full</b>     | raw, 3      | ridge shared / horizon | ridge / convex shared | ridge / delta shared |
| <b>cov_avgy</b> | instance, 3 | ridge shared / horizon | ridge / convex shared | ridge / delta shared |

Each row retains **<stem>\_ridge\_shared** as its control. The three fronts replace only the suffix by **ridge\_horizon**, **convex\_shared**, or **delta\_ridge\_shared**, respectively.

Table 14: Selected CatBoost controls and one-axis transformations.

| $A$         | Space, $K$  | Control          | Separate one-axis variants                                  |
|-------------|-------------|------------------|-------------------------------------------------------------|
| <b>avgy</b> | instance, 1 | regressor shared | classifier shared; regressor shared soft; regressor horizon |
| <b>cov</b>  | raw, 1      | regressor shared | classifier shared; regressor shared soft; regressor horizon |

For each  $A$ , the CatBoost front also includes matching direct, Bayes, and oracle references. No transformation is crossed with another.

Table 15: Root submission fronts and tested model sets.

| Front                                         | Model set / controlled axis                                                                                              |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>screen.slurm</code>                     | $\mathcal{B}_{\text{screen}}$ and $\mathcal{G}_{\text{screen}}$ on $\mathcal{D}_P \times \mathcal{S} \times \mathcal{R}$ |
| <code>benchmark.slurm</code>                  | complete selected screen pipelines; <b>full</b> uses Chronos-2, <b>ultra</b> adds TabPFN-TS                              |
| <code>k_ablation.slurm</code>                 | selected pipelines, $K \in \{1, 3, 5, 10, 15, 20\}$ only                                                                 |
| <code>h_ablation.slurm</code>                 | selected pipelines, $L = 504$ , $H \in \{24, 168, 504\}$ only                                                            |
| <code>l_ablation.slurm</code>                 | selected pipelines, $H = 24$ , $L \in \{24, 168, 504\}$ only                                                             |
| <code>horizon_baselines_ablation.slurm</code> | five controls in Table 13; shared versus horizon ridge                                                                   |
| <code>convex_baselines_ablation.slurm</code>  | same five; shared ridge versus shared convex                                                                             |
| <code>delta_baselines_ablation.slurm</code>   | same five; shared ridge versus shared delta-ridge                                                                        |
| <code>catboost_ablation.slurm</code>          | two controls in Table 14; three separate one-axis variants                                                               |
| <code>mixed_quantity_ablation.slurm</code>    | selected pipelines transferred to the five mixed-quantity panels                                                         |
| <code>fourier_retrieval_ablation.slurm</code> | selected pipelines under their original and Fourier-amplitude retrieval spaces                                           |
| <code>offline_datastore_ablation.slurm</code> | selected pipelines under online and fixed T0-only retrieval                                                              |
| <code>retrieval_scope_ablation.slurm</code>   | selected pipelines with all-user, same-user-only, and other-users-only retrieval                                         |
| <code>sota_benchmark.slurm</code>             | selected Chronos-Bolt method on exact paper splits plus immutable published TS-RAG/Cross-RAG rows                        |
| <code>tsrag.slurm</code>                      | released TS-RAG and the selected project method on matching fixed $K = 10$ neighbours                                    |
| <code>ts_ifa.slurm</code>                     | complete non-meta routing/branch grid over the selected test/full/ultra subset                                           |
| <code>ts_ifa\h_ablation.slurm</code>          | selected candidates at $L = 504$ over three horizons                                                                     |
| <code>ts_ifa\l_ablation.slurm</code>          | selected candidates at $H = 24$ over three lookbacks                                                                     |
| <code>ts_ifa\meta\ridge.slurm</code>          | meta-learning on textually selected ridge candidates                                                                     |
| <code>ts_ifa\meta\neural.slurm</code>         | meta-learning on textually selected neural candidates                                                                    |
| <code>extraction.slurm</code>                 | standalone extraction smoke or TS-IFA-input front                                                                        |
| <code>baselines.slurm</code>                  | standalone primary-baseline smoke front                                                                                  |
| <code>gates.slurm</code>                      | standalone primary-gate smoke front                                                                                      |
| <code>tables.slurm</code>                     | standalone result-table front                                                                                            |

Manual follow-up selection has two immutable generations. The root `SWEEP_CANDIDATES.txt` records the screen-selected pipelines used by the already-launched formulation, retrieval, K/H/L, and mixed-panel studies. After the K study is complete and analyzed, promoted best-K adaptation identities are written to `SECOND_GENERATION_CANDIDATES.txt`; this second manifest alone feeds the final benchmark, TS-RAG, and SOTA fronts. Selected TS-IFA identities are appended to it only after the routing pilot is analyzed, before the TS-IFA H/L and meta follow-ups. Non-comment records in both files are complete **family/retrieval/method** identities, and the shared reader filters the applicable adaptation, shared-ridge, shared-CatBoost, TS-IFA ridge, or TS-IFA neural entries for each front. `SELECTED_CANDIDATES_FILE` may select another manifest; front-specific CSV variables remain explicit one-launch overrides. A consumer fails clearly while its required generation-2 entries are absent.

The corrected current screen ranking compares each adapted result only with the vanilla result from the same retrieval run. The frozen first-generation manifest holds the top five shared-ridge pipelines, the best shared CatBoost cov/avgY pipeline, and the matching shared Bayes references. The completed  $K$  study promotes eight unique generation-2 identities in decreasing order of their best- $K$  held-out improvement; the two first-generation raw-full rows collapse to one identity. The first record is `baselines/instance_euclidean_5_online/full_ridge_shared`, so the final benchmark evaluates all eight records while TS-RAG and SOTA use that leading shared-ridge control. Bayes ties retain the smallest  $K = 1$ . No convex or delta baseline was eligible for the screen. Earlier formulation and  $K$  summaries derived from the old cross-retrieval ranking remain invalid; the current reports below were regenerated through the corrected reader. Publication CatBoost fits allow 300 trees; only the smoke profile uses two.

### 9.3 Current held-out results

All results in this subsection are complete held-out  $T_3$  comparisons against the vanilla forecast from the same retrieval run. Improvements are the equal-configuration mean  $I_{m,c}$  defined below; window wins are the mean of  $W_{m,c}$ . The screen contains 12 dataset-setting configurations.

Table 16: Leading pipelines from the corrected shared-method screen. The five ridge rows are the selected baseline sweep; the final row is the strongest deployable learned gate.

| Method                        | Retrieval          | $K$ | Mean $I_{m,c}$ | Window wins   |
|-------------------------------|--------------------|-----|----------------|---------------|
| full_ridge_shared             | instance Euclidean | 3   | <b>+1.95%</b>  | 53.18%        |
| full_ridge_shared             | raw Euclidean      | 1   | +1.79%         | 53.40%        |
| cov_y_ridge_shared            | instance Euclidean | 3   | +1.78%         | 53.65%        |
| full_ridge_shared             | raw Euclidean      | 3   | +1.78%         | 52.64%        |
| cov_avgy_ridge_shared         | instance Euclidean | 3   | +1.76%         | <b>54.17%</b> |
| catboost_cov_regressor_shared | raw Euclidean      | 1   | +0.03%         | 3.50%         |

The baseline sweep therefore selects shared ridge, not a gate. The full design with instance-normalized retrieval and  $K = 3$  gives the largest mean reduction, while all five selected ridge pipelines lie in the narrow +1.76–1.95% band. The CatBoost gate is effectively tied with vanilla and improves too few windows to be a competitive primary adaptor.

Table 17: Matched one-axis retrieval and formulation ablations. Each entry is mean held-out nMSE improvement over its retrieval-matched vanilla run. The shared column is the unchanged control in the dedicated ablation reports.

| Control pipeline            | Shared       | Fourier | Horizon | Convex | Delta |
|-----------------------------|--------------|---------|---------|--------|-------|
| full, instance, $K = 3$     | <b>+1.95</b> | +1.33   | +1.52   | −0.05  | +0.61 |
| full, raw, $K = 1$          | <b>+1.79</b> | +1.28   | +1.33   | +0.55  | +0.57 |
| cov_y, instance, $K = 3$    | <b>+1.78</b> | +1.37   | +1.35   | +0.71  | +0.47 |
| full, raw, $K = 3$          | <b>+1.87</b> | +1.33   | +1.45   | +0.88  | +0.82 |
| cov_avgy, instance, $K = 3$ | <b>+1.76</b> | +1.62   | +1.43   | +0.76  | +0.54 |

The original Euclidean retrieval beats the Fourier-amplitude alternative for every selected pipeline. Shared ridge also beats its matched per-horizon, convex, and delta-ridge replacement in every row. The extra horizon parameters do not compensate for their higher variance, and the convex and delta constraints remove useful signed fusion freedom. These alternatives remain ablation-only; shared ridge is the sole retained baseline formulation.

Table 18: Completed baseline lookback ablation at fixed  $H = 24$ . Each cell reports mean held-out nMSE improvement / mean positive-window rate (%), averaged equally over Electricity, Traffic, Solar, and Exchange.

| Pipeline                    | $L = 24$      | $L = 168$            | $L = 504$            |
|-----------------------------|---------------|----------------------|----------------------|
| full, raw, $K = 3$          | +6.72 / 55.34 | +1.04 / 49.70        | +0.24 / 48.78        |
| full, raw, $K = 1$          | +4.49 / 51.89 | +0.81 / 50.05        | +0.37 / 49.14        |
| full, instance, $K = 3$     | +2.43 / 53.03 | <b>+1.14</b> / 49.85 | +0.41 / 48.64        |
| cov_y, instance, $K = 3$    | +2.90 / 53.05 | +0.90 / 50.40        | +0.97 / 49.02        |
| cov_avgy, instance, $K = 3$ | +2.92 / 52.35 | +0.93 / 50.65        | <b>+1.03</b> / 49.50 |

Baseline lookback-length ablation (equal mean over four datasets)

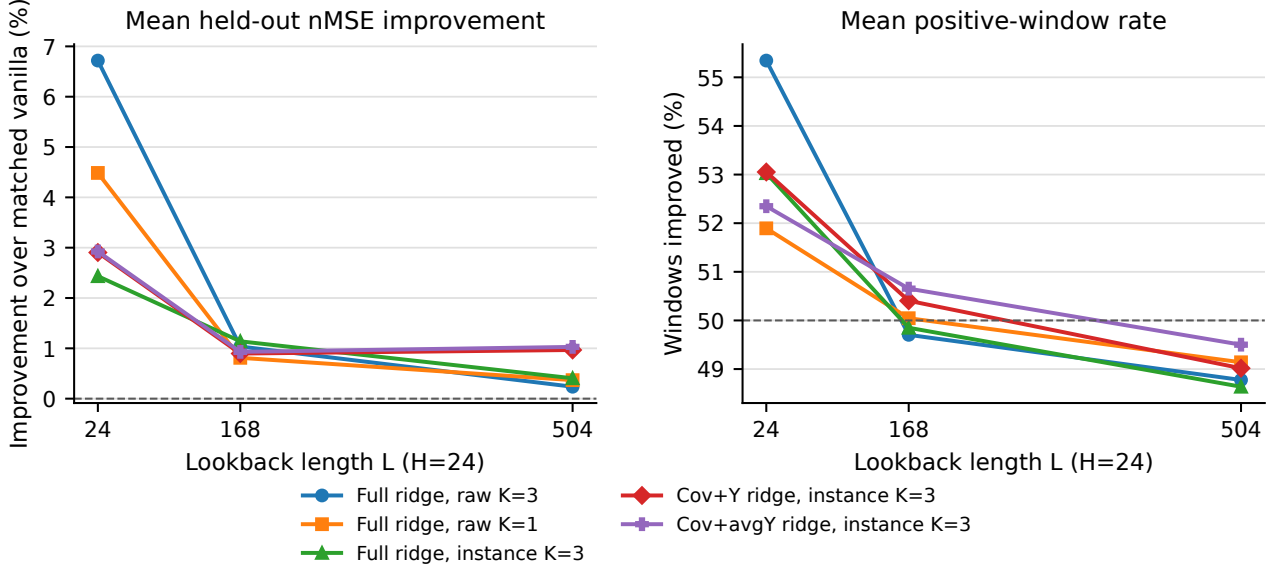


Figure 1: Baseline lookback-length ablation. Dashed references mark no nMSE improvement and a 50% window win rate.

Table 19: Lookback and horizon sensitivity of the screened `full_ridge_shared` instance- $K = 3$  control. Entries are equal-dataset mean held-out nMSE improvements (%) over the retrieval-matched vanilla forecast.

| Varied axis  | Fixed axis | 24    | 168          | 504   |
|--------------|------------|-------|--------------|-------|
| Lookback $L$ | $H = 24$   | +2.43 | +1.14        | +0.41 |
| Horizon $H$  | $L = 504$  | +0.41 | <b>+2.78</b> | +2.75 |

The apparent  $L = 24$  maximum is not stable: its dataset effects range from  $-7.37\%$  on Exchange to  $+16.48\%$  on Solar, while the Traffic vanilla nMSE is  $2.07 \times 10^{10}$ . We therefore retain  $L = 168$  as the balanced short-horizon setting. At fixed  $L = 504$ , the gain rises from  $+0.41\%$  at  $H = 24$  to  $+2.78\%$  at  $H = 168$  and then plateaus at  $+2.75\%$  for  $H = 504$ , showing that the adaptor’s relative benefit persists at long horizons.

Table 20: Transfer of the best screened `full_ridge_shared` instance- $K = 3$  pipeline to panels whose channels mix physical quantities. The primary row is the homogeneous screen reference; mixed rows average three  $(L, H)$  settings.

| Panel                      | Mean $I_{m,c}$ | Window wins |
|----------------------------|----------------|-------------|
| Primary homogeneous screen | <b>+1.95%</b>  | 53.18%      |
| ETTh1 mixed                | +3.08%         | 52.88%      |
| ETTh2 mixed                | +1.19%         | 49.98%      |
| ETTM1 mixed                | +8.60%         | 52.58%      |
| ETTM2 mixed                | +0.91%         | 53.59%      |
| Weather mixed              | -125.87%       | 42.78%      |
| All five mixed panels      | -22.42%        | 50.36%      |

Some ETT panels remain locally adaptable, but mixing unrelated variables makes the method non-robust: Weather reverses the aggregate result, and raw-retrieval full designs additionally exhibit catastrophic ETTm1 outliers. Cross-variable retrieval and fusion should therefore be restricted to channels representing the same physical quantity. Mixed-quantity panels are diagnostic ablations, not members of the primary benchmark.

## 9.4 Result identity, repetitions, and selection

Extraction is a reusable workflow with identity

outputs/extraction/dataset/L\_H/backbone/space/metric/k/mode/run\_n/

and vanilla uses `none/none/0/none`. Baselines, gates, Cross-RAG, and each ablation use a separate family root with identity

outputs/adaptation/family/dataset/L\_H/backbone/formula/space/metric/k/mode/run\_n/

TS-IFA instead orders `variant/routing_scope/routing_constraint/branch_set` before `space/metric/k/mode`. Each config owns one directory. Optional row and column config subsets affect only table layout.

Splits, budgets, optimizers, regularizers, fit caps, candidate lists, and evaluation controls are pipeline configs in `manifest.json`; device and scheduler placement are runtime-only. The recorded seed fixes all stochastic choices. Repeated seeds, when used, are `seed_n` leaves. Mode is only a test/full/ultra path-subset selector and is never a family, phase, path field, or computation-signature field.

Downstream allocation resolves the extraction dependency by its complete expected pipeline configuration and exact seed and fails unless precisely one run remains. Its signature embeds the upstream schema, identity, pipeline and experiment configuration, and seeds. The upstream path and manifest ID remain plain provenance and never affect reuse. Thus a smoke extraction cannot satisfy a publication computation merely because both occupy the same identity path.

Every run uses `schema_version: 1`, declared configs, optional plain provenance, launch attempts, required artifacts, and status not-run/running/interrupted/completed. Run identity contains only the schema, identity/model configs, pipeline/experiment configs, and seeds. It never fingerprints or hashes code, Slurm files, data, weights, logs, outputs, checkpoints, or directories; code/data changes are manual rerun decisions. The schema changes only for a deliberate global contract break, and `new` creates another repeat with unchanged parameters. The default `overwrite_exact` collision policy skips exact completion, resumes exact interruption, and adds `run_n` for a changed pipeline. Before a non-skipped attempt becomes running, schema-owned preparation removes stale generated artifacts while preserving `manifest.json`, archived manifest history, and completed `seed_n` leaves. Computation entry points never delete the allocated `run_n` root. A manifest-less `run_n` is an invalid partial directory and is reclaimed by the next allocation at that index. The overall run remains `running` with `ready_at_utc` while finished seed state is `ready`; completion is recorded immediately after that producer's `srun` returns successfully. A later configuration or table failure preserves completed producers and interrupts only unfinished work. Completed manifests are authoritative and are not followed by file hashing or synchronization checks. Within that active workflow, later stages may select ready manifests only when their launch ID matches; independent reuse remains completed-only. Tables support distinct/latest/average pipeline configurations and selected/latest/distinct/average exact repeats. Explicit pipeline filters select configurations and must match even with one run. `SELECTED_RUNS.txt` records only automatic or pinned exact repeats, and every report records requested filters and obtained manifests. Unmigratable legacy material is isolated under `archive/legacy_pre_schema_v1_2026-08-07` and is never read by current tables.

Slurm workflows never submit a publisher or execute Git commands. After any terminal state, including failure, cancellation, or timeout, the user runs `bash publish_job.sh <job-id>` from the project root. The script verifies `main`, sources `$HOME/codes/proxy.sh` (or `PROXY_SCRIPT_PATH`), and fast-forward pulls `origin/main` before artifact selection or staging. With a job ID, it commits only the exact stdout/stderr pair. Omitting the job ID publishes the complete log and lightweight output trees, excluding `*.pt`, `*.npz`, and `*.cbm`. It then pushes `origin/main` without creating a pull request. A non-fast-forward pull stops without creating a merge commit; unrelated staged paths remain outside the publisher commit.

Table construction accepts only current result manifests, keeps selected pipelines family-qualified, and requires complete dataset-setting-model coverage before writing any table. It also reads every selected current `baseline_artifacts.pt`, expands the neighbour coefficients by rank, and writes exact signed coefficient CSV files and imshow heatmaps; shared models have one row and horizon models have one row per horizon. Direct baselines have no fitted coefficients and are omitted. Complete extraction, baseline, and gate configurations are reused by default, so resubmitting a profile or ablation front rebuilds these tables and plots without refitting. For the  $K$  ablation, the same stage also overlays every candidate's average held-out nMSE improvement against  $K$  and marks the global optimum. The obsolete combined TS-IFA directory is not a valid input; each current TS-IFA method owns its isolated manifest and result folder.

For setting  $c$ , method  $m$ , method score  $\text{nMSE}_{m,c}$ , and matching vanilla  $\text{nMSE } v_c$ , the reported percentage improvement is

$$I_{m,c} = 100 \frac{v_c - \text{nMSE}_{m,c}}{v_c}.$$

The average table reports the unweighted mean of  $I_{m,c}$  over complete settings, not a percentage computed after pooling losses. If  $n_c$  is the number of held-out windows in setting  $c$ , window-level win rate is

$$W_{m,c} = 100 \frac{1}{n_c} \sum_{i=1}^{n_c} \mathbf{1} \left[ \frac{1}{H} \sum_{h=1}^H (G_{i,h} - \hat{G}_{m,i,h})^2 < \frac{1}{H} \sum_{h=1}^H (G_{i,h} - V_{i,h})^2 \right].$$

Final claims use complete held-out  $T_3$  scoring. Optimistic `*_eval_fit` baselines and target-aware oracle gates are appendix diagnostics only.

## 10 Summary

The project tests a hierarchy of increasing adaptation capacity around frozen forecasting foundation models: direct analogue forecasts, ridge/convex/delta linear families, conditional CatBoost gates, and TS-IFA. Every method shares the same query dates, leakage-free online datastore, query-scale neighbour signals, and held-out  $T_3$  evaluation. This makes capacity, retrieval choice, and validation protocol explicit rather than conflating them with backbone quality. Current complete evidence selects the shared full ridge baseline with instance-normalized Euclidean retrieval and  $K = 5$ ; its completed  $K$  sweep gives a +2.00% equal-configuration mean nMSE improvement. Fourier retrieval and per-horizon, convex, or delta formulations are uniformly weaker in matched comparisons, while mixed-quantity panels are unstable enough to justify homogeneous-quantity grouping as the primary contract.

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